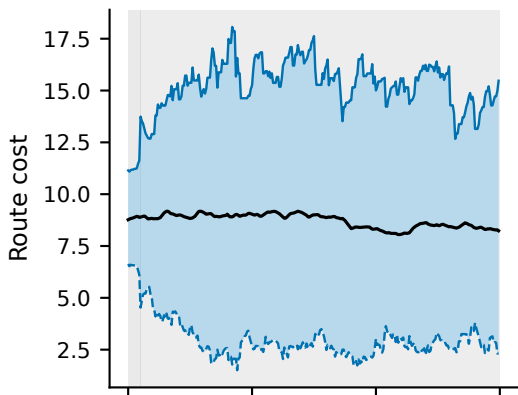


warm-up
 certified
 [LB, UB]
 UB
 valid
 true OPT
 LB



warm-up
 gap (UB-LB)
 $\epsilon = 5$
 valid (uncertified)

